GOLDLITE SUPPLIES

KCBE GRADE 10 QUICK REVISION SERIES 2026

PHYSICS

SERIES 1-10

SERIES 1

SECTION A (30 MARKS)

Answer ALL questions in this section.

1. Three learners at **Mang'u High School** are discussing the relevance of Physics.

- **Abdi:** "I want to study how stars and planets move."
- **Bakari:** "I am interested in how heat energy moves in car engines."
- **Chantal:** "I want to design circuits for the next Kenyan-made smartphone."

(a) **Identify** the branch of Physics each learner is interested in: (3 marks)

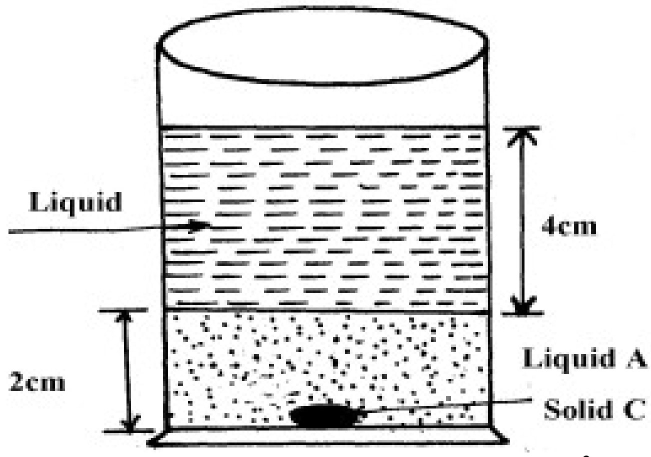
i. Abdi: ____________________________.
ii. Bakari: ____________________________.
iii. Chantal: ___________________________.

(b) **Outline** TWO importance of Physics in the field of Medicine. (2 marks)

i. ____

ii. ____

2. Two immiscible liquids are poured in a container to the levels shown in the diagram below.



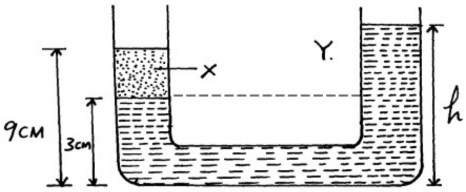
If the densities of the liquids A and B are $1 g/cm^3$ and $0.8 g/cm^3$ respectively, find the pressure acting upon solid C at the bottom of the container due to the liquids

(b) **Name** the two factors that affect pressure at a point in a liquid. (2 marks)

i. ____

ii. ____

3. In the diagram below, the U-tube contains two liquids; X and Y which do not mix. If the density of liquid Y is $900 Kgm^{-3}$ and that of X is $1200 Kgm^{-3}$, calculate the height of liquid Y



4. Grade 10 learners at **Kisumu Boys** are testing a copper wire and a glass rod.

(a) **Identify** the property of the material represented by the curve that snaps suddenly without stretching. (1 mark)

(b) **Define** *Hooke's Law*. (2 marks)

(c) **Distinguish** between *Ductility* and *Malleability*. (2 marks)

5.

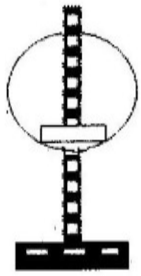
a.

i. State coulombs law of electrostatic force

ii. Define capacitance

b. Describe how the type of charge on a charged metal rod can be determined

c. The fig. Shows hollow negatively charged sphere with a metal disk attached to an insulator placed inside. State what would happen to the leaf of an uncharged electroscope if the metal disk were brought near the cap of the electroscope. Give a reason for your answer.



____

____

____

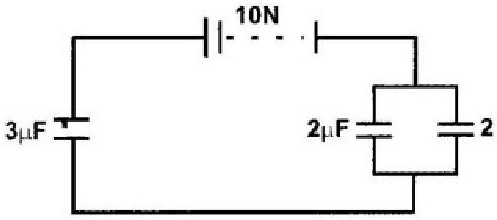
____

d. State two ways of charging the magnitude of the deflection of the leaf of an electroscope.

i. ____

ii. ____

e. The fig below shows an arrangement of capacitors connected to a 10V d.c supply.



Determine:

i. The charge stored in the 2μF capacitor.

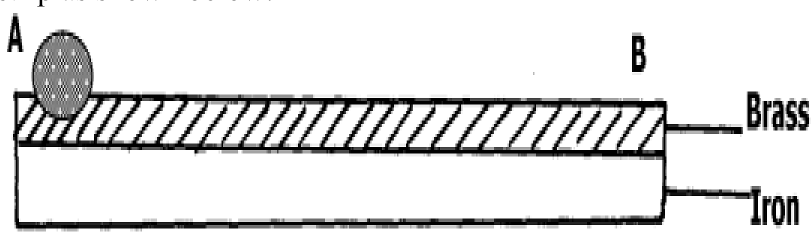
ii. The total capacitance of the arrangement.

iii. **Explain** the process of *Earthing*. (2 marks)

____

____

6. The figure below shows a bimetallic strip made of brass and iron. A marble is placed at end A of the bimetallic strip as shown below:-



(a) State and explain what will be observed when the bimetallic strip is strongly cooled

____

____

(b) **Identify** which metal expands more between Brass and Iron based on the curve. (1 mark)

____

(c) **Mention** two applications of bimetallic strips in home appliances. (2 marks)

i. ____

ii. ____

(d) **State** the SI unit of Temperature.________________________________ (1 mark)

(e) **True or False:**

Water contracts when cooled from 4∘C to 0∘C. [__________] (1 mark)

7. (a) **Identify** the material type (Insulator, Conductor, or Semiconductor) if the forbidden gap is very large. (1 mark)

(b) **Name** the process of adding impurities to an intrinsic semiconductor. (1 mark)

____

(c) **Distinguish** between an **n-type** and a **p-type** semiconductor. (3 marks)

N-type	**p-type**

SECTION B: [50 MARKS]

Answer ALL questions in the spaces provided.

8. A bus company in **Eldoret** is designing a new double-decker bus. The engineers must ensure it does not topple on sharp corners.

(a) **Define** *Centre of Gravity (C.O.G)*. (2 marks)

____

____

(b) **Explain** why the bus with luggage on the roof is less stable than the empty one. (3 marks)

i. ____

ii. ____

iii. ____

(c) **State** the two conditions for a body to be in static equilibrium. (4 marks)

i. ____

ii. ____

(d) A uniform meter rule is pivoted at the 50cm mark. A 2N weight is placed at the 10cm mark. **Calculate** where a 5N weight should be placed to balance the rule. (3 marks)

2. Construction workers in **Kilifi** are using a pulley system to lift bags of cement to the third floor.

a. Define the term mechanical advantage of a machine. (1mk)

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b The figure shows a pulley system being used to raise a load.

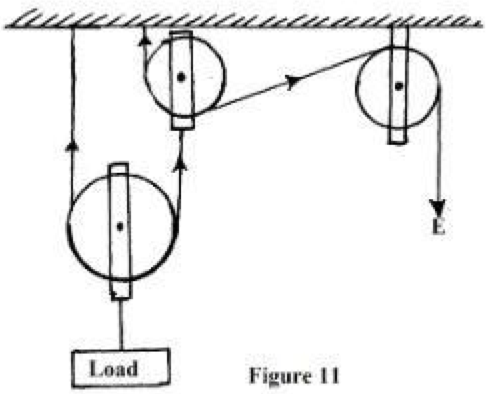


Figure 11

This pulley system has an efficiency of 75%.

i. Determine the velocity ratio of the system. (1mk)

ii. Calculate the mechanical advantage of the pulley system. (2mks)

iii What effort is required to raise a load of 240 kg? (2mks)

iv. Calculate the work done by a person using this machine in raising a load of 120 kg through a vertical distance of 2.5 m (3mks)

c. Give two reasons to explain why the efficiency of a machine cannot be 100%. (2mks)

i. ______________________________

ii. ______________________________

d. **Define** *Efficiency* of a machine. (2 marks)

---

e. **Explain** why the efficiency of this pulley system can never be 100%. (3 marks)

i. ______________________________

ii. ______________________________

f. A worker lifts a 50kg bag through a height of 10m in 20 seconds. **Calculate**:

i. Work done (2 marks)

ii. Power developed (2 marks)

g. **Name** three other types of simple machines used in construction. (3 marks)

i. ______________________________

ii. ______________________________

iii. ______________________________

3. An ambulance is moving towards a hospital in **Machakos** with its siren on.

(a) **Name** the effect described where the frequency of sound changes due to relative motion. (2 marks)

____

____

(b) **Describe** how the pitch heard by the observer changes as the ambulance approaches. (3 marks)

____

____

(c) **Identify** the wave property shown in the diagram below where waves bend around a corner. (2 marks)

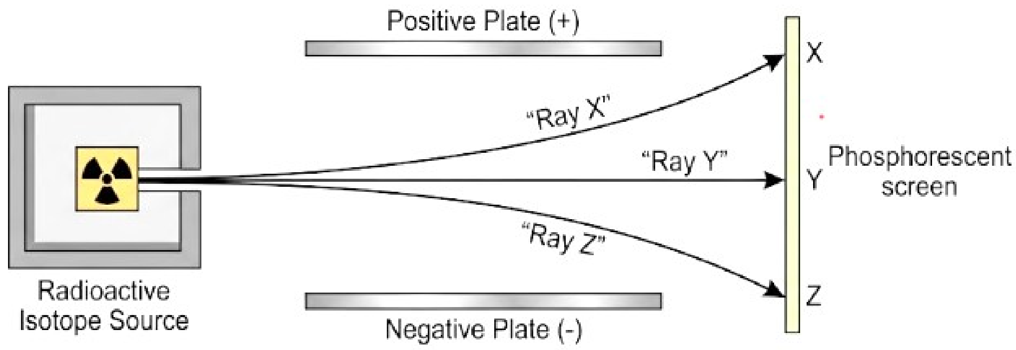
____

____

(d) **Distinguish** between *Refraction* and *Reflection* of waves. (5 marks)

Refraction	Reflection

4. Doctors at **Kenyatta National Hospital** use radioactive isotopes for cancer treatment, while environmentalists discuss the **Greenhouse Effect**.



(a) **Identify** the three types of radiations emitted during radioactive decay from the diagram. (3 marks)

Ray X: ________________

Ray Y: ________________

Ray Z: ________________

(b) **State** two safety precautions when handling radioactive materials. (2 marks)

i. ____

ii. ____

(c) **Explain** how the **Greenhouse Effect** leads to **Global Warming**. (4 marks)

i. ____

ii. ____

iii. ____

iv. ____

(d) **Mention** two human activities that contribute to the depletion of the **Ozone Layer**. (2 marks)

i. ____

ii. ____

(e) **Define** *Half-life*. (2 marks)

____

____

SERIES 2

SECTION A: (30 marks)

Answer ALL questions in this section.

1. **Allan** a grade 10 learner at **Nyagatugu Senior School** is taking physics and Computer combination, Define physics as a learning area. (2 marks)

____

____

2. **Amina** works as a doctor at **Sunrise Medical Centre**. How do you think Physics has helped make her work easier as a doctor? (2 marks)

____

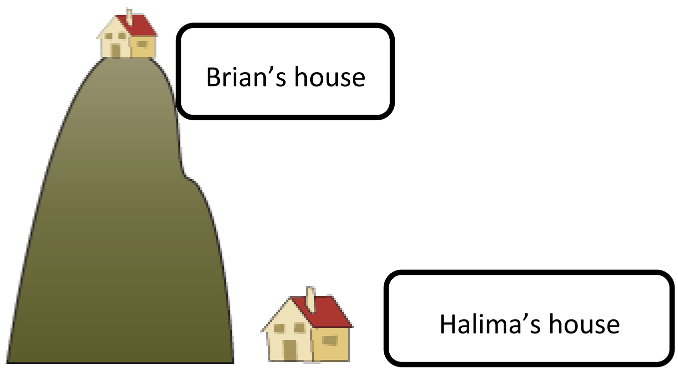
____

3. **Physics** and **Computer Science** are related. Explain this interrelationship. (2 marks)

____

____

4. When doing his research, **Ochieng** found out that Physics has so many branches. Name the branch that: (3 marks)
   a) Focuses on electric and magnetic fields and how they interact. ____________________
   b) Studies the motion of objects and the forces acting on them. ____________________
   c) Studies of objects in space and the universe as a whole. ____________________
5. Grade 10 learners, **Brian and Halima**, found out that atmospheric pressure changes with altitude. So, they decided to set up a house at different altitudes as shown below.

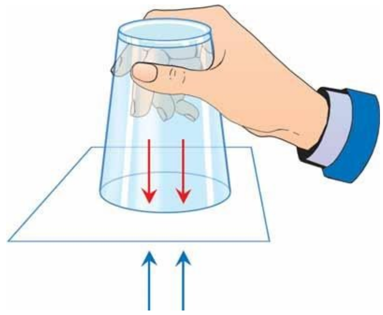


Whose house stood at a place with high atmospheric pressure? (1 mark)

____

____

6. During a Physics experiment, **Mutiso** conducted an experiment using glass full of water and a piece of paper as shown below.



In relation to atmospheric pressure, explain why the water did not four upon inverting the glass. (2 marks)

____

____

7. How does the density of a fluid affect pressure in fluids? (2 marks)

____

____

8. A container is filled with mercury whose density is 13,600 kg/m^3 to a height of 0.8 m. Find the pressure at the bottom of the container. (Take g = 9.8 N/kg). (2 marks)

9. During a Physics lesson Grade 10 learners were guided to understand the principle of transmission of pressure in fluids. State the Pascals' principle. (2 marks)

____

____

10. **Njeri** wants to measure the temperature of her class. Name two instruments that she can use to measure this temperature. (2 marks)

____

____

11. Mr. **Otieno** has a steel rod measures 75.0 cm at 20°C. When heated to 120°C, its length becomes 75.006 cm. Calculate the linear expansivity of steel. (3 marks)

12. Categorize the following states of equilibrium as either stable equilibrium, unstable equilibrium or neutral equilibrium. (3 marks)

a) ____________________ b) ____________________ c) ____________________

13. State a reason why brake fluid is the most preferred fluid in braking systems instead of any other liquid. (1 mark)

____

____

14. **Kevin Mwangi** claims that springs are elastic. What does this mean? (1 mark)

____

____

15. According to the books of history, the highest temperature to be recorded on earth is 56°C and was recorded in USA in 1913. Convert this temperature to kelvin. (2 marks)

____

____

SECTION B (50 marks)

Answer all the questions in the spaces provided after each question.

16. Learners discussed the importance of Physics in daily life.

a) State two ways Physics supports communication and technology. (2 marks)

i. ____________________.
ii. ____________________.

b) Outline three ways Physics contributes to transportation and safety. (3 marks)

i. ____________________.
ii. ____________________.
iii. ____________________.

c) State four natural phenomena explained using Physics. (4 marks)

i. ____________________.
ii. ____________________.
iii. ____________________.
iv. ____________________.

17. During a class activity, learners converted temperature between Celsius and Kelvin scales.

a) State the SI unit of temperature. (2 marks)

b) Convert the following temperatures: (6 marks)
i) 25°C to kelvin

ii) 300 K to degrees Celsius

iii) 15°C to kelvin

c) State two precautions taken when using a liquid-in-glass thermometer. (2 marks)

i. ______.
ii. ______.

18. Explain four applications of expansion and contraction of matter in day-to-day life. (8 marks)

i. ______.
ii. ______.
iii. ______.
iv. ______.

19. During a Physics lesson, learners were introduced to atmospheric pressure.

a) What is atmospheric pressure? (2 marks)

______
______

b) Explain why atmospheric pressure decreases with altitude. (2 marks)

i. ______.
ii. ______

20. Explain how atmospheric pressure is applied in the following instances.

a) Using a syringe to draw water (3 marks)

i. ______.
ii. ______.
iii. ______.

b) Drinking through a straw (3 marks)

i. ____.
ii. ____.
iii. ____.

c) Siphoning (3 marks)

i. ____.
ii. ____.
iii. ____.

21. Below are some mechanical properties of materials. Describe them. (10 marks)

	Property	**Description/ definition**
i	Ductility	
ii	Malleability	
iii	Brittleness	
iv	Stiffness	
v	Toughness	

SERIES 3

SECTION A: (100 Marks)

Answer all questions in this section.

1. a) Define **Physics** as a body of knowledge in science. (2 marks)

____

____

b) Identify **FOUR branches of Physics**. (4 marks)

i.____________________________. ii.____________________________.

iii.____________________________. iv.____________________________.

c) Explain **THREE ways Physics is important in day-to-day life**. (3 marks)

i.____.

ii.___________________________________ ___________________________________.

iii.__________________________________ ___________________________________.

d) Describe **ways Physics is related to other fields of study**. (3 marks)

i. Physics and Biology

.____.

.___________________________________ ___________________________________.

ii. Physics and Chemistry

.____.

.___________________________________ ___________________________________.

iii. Physics and Geography

.____.

.___________________________________ ___________________________________.

e) State **three career opportunity** associated with Physics. (3 marks)

i.____.

ii.___________________________________ ___________________________________.

iii.__________________________________ ___________________________________.

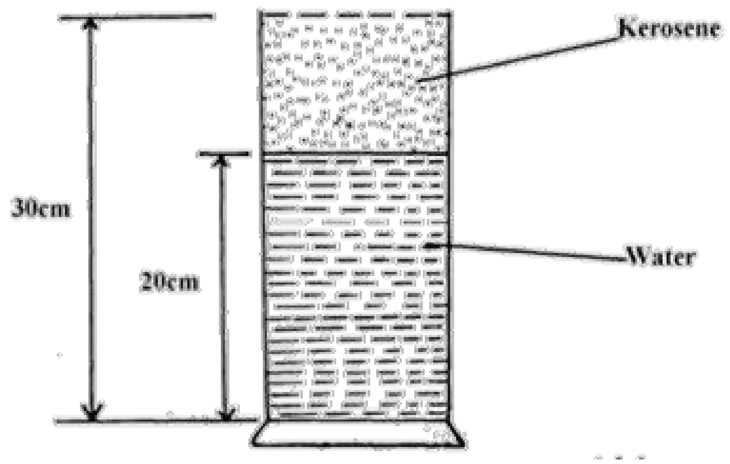
2. State **THREE factors that affect pressure in liquids**. (3 marks)

i.____.

ii.____.

iii.____.

3. The figure below shows a measuring cylinder of height 30cm filled to a height of 20cm with water and the rest occupied by kerosene



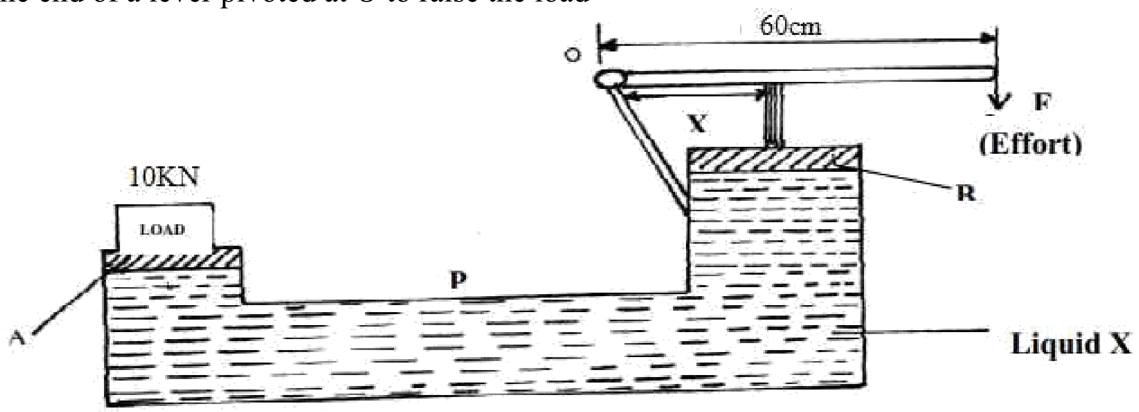
Given that density of water = 1000Kgm^{-3}, density of kerosene = 800Kgm-3 and atmospheric pressure = 1.03x10^{5} pascals, determine the pressure acting on the base of the container. (3 marks)

4. State Pascal's principle of transmission of pressure(3 marks)

____

____

5. The figure below shows a hydraulic press P which is used to raise a load of 10KN. A force F of 25N is applied at the end of a lever pivoted at O to raise the load



a. State one property of liquid X (1 mark)

i.____.

b. Determine the distance x indicated on the press if force on piston B is 100N. (3 marks)

6. *A copper wire being stretched by a Grade 10 learner.*

a) Define the following properties of materials: (4 marks)
i) Ductility

____

____

ii) Elasticity

____

____

b) State **TWO materials** that show **malleability** in everyday life. (2 marks)

i.____.

ii.____.

c) A wire of original length **2.0 m** extends by **0.004 m** when a force is applied. (3 marks)
Calculate the **strain**. (3 marks)

d) A force of **400 N** is applied on a wire with cross-sectional area **0.0002 m²**.
Calculate the **stress**. (3 marks)

e) State **TWO industrial applications** of mechanical properties of materials. (2 marks)

i.____.

ii.____.

f) Explain why **steel** is preferred over **glass** in construction. (2 marks)

i.____.

ii.____.

7. State the reason why the speed of water at the narrow tube/section of a river is higher than at the wider section. (2mks)

____.

____.

____.

8. State Hooke's law of elasticity (1mark)

____.

______________________________.

______________________________.

9. a) Define **temperature**. (2 marks)

____

____

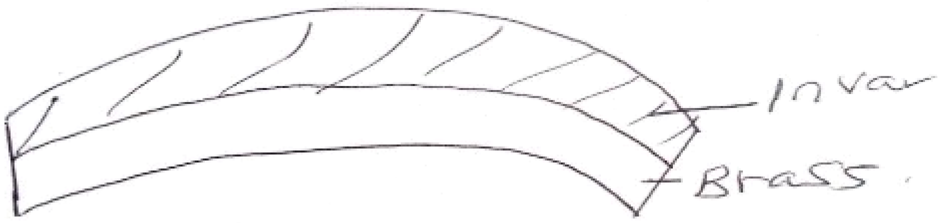
b) Name **THREE units of temperature**. (3 marks)

i.______________________________.

ii.______________________________.

iii.______________________________.

c) The figure below shows the shape of a bi-metallic strip after it has cooled below room temperature.



Explain why the strip curved as shown. (3marks)

.______________________________.

.______________________________.

.______________________________.

d) Identify **THREE temperature-measuring devices**. (3 marks)

i.______________________________.

ii.______________________________.

iii.______________________________.

e) Describe the **unusual expansion of water** between 0°C and 4°C. (2 marks)

.______________________________.

.______________________________.

.______________________________.

f) State **THREE applications of thermal expansion** in daily life. (3 marks)

i.______________________________.

ii.____ ____.

iii.____ ____.

10. a) Define: (4 marks)
i) Moment of a force

.____.

.____ ____.

ii) Torque
____.

.____ ____.

b) State the **principle of moments**. (2 marks)

.____.

.____ ____.

.____ ____.

c) Explain **TWO states of equilibrium**. (4 marks)

i.____.

ii.____ ____.

e) Describe how the **centre of gravity** affects the stability of an object. (3 marks)

i.____.

ii.____ ____.

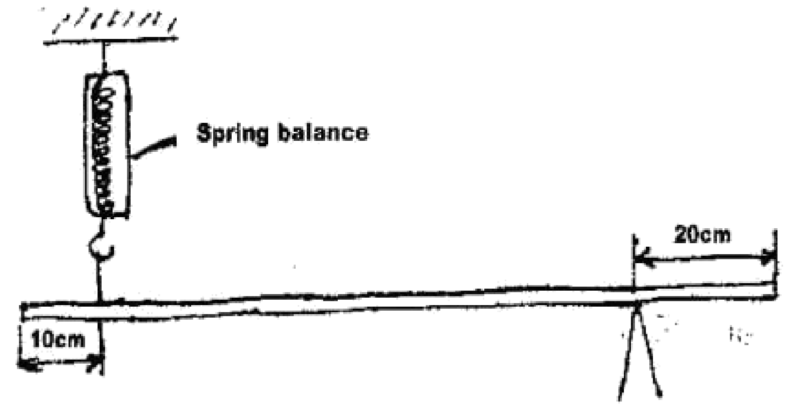
iii.____ ____.

f) State **TWO applications of moments** in daily life. (2 marks)

i.____.

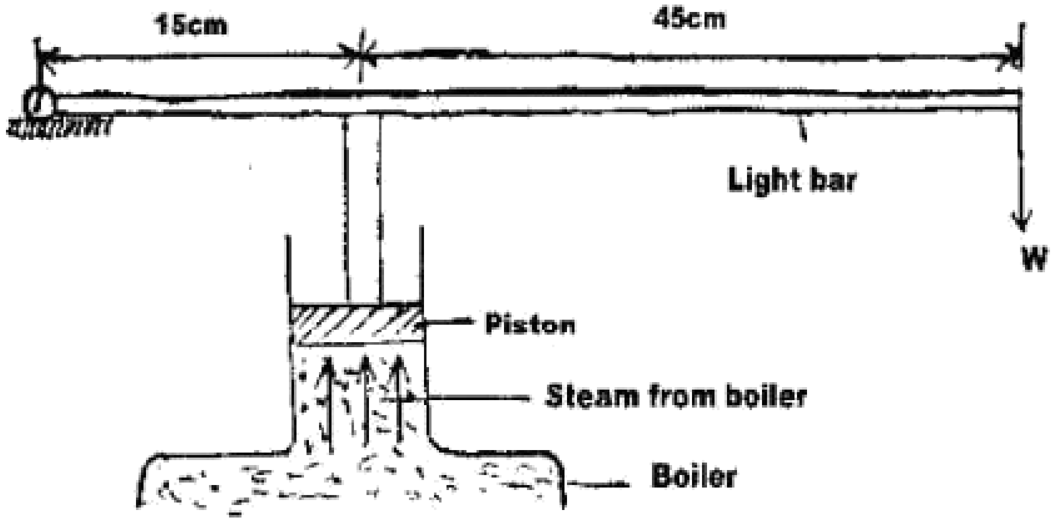
ii.____ ____.

11. The figure below shows a uniform bar of length 1 m pivoted near one end. The bar is kept in equilibrium by a spring balance as shown.



Given that the reading of the spring balance is 0.6N. Determine the weight of the bar. (3 marks)

12. The figure shows a device for closing a steam outlet.



The area of the piston is 4.0×10^{-4} m^2 and the pressure of the steam in the boiler is 2.0×10^5 Nm^{-2}. Determine the weight W that just holds the bar in the horizontal position shown. (3 marks)

13.

a. State the principle of moments.
b. Two men P and Q carried a uniform ladder 3.6 m long weighing 1200N. P held the ladder from one end while Q supported the ladder at a point 0.4m from the other end.
    i. Sketch a diagram showing the forces acting on the ladder. (3 marks)

    ii. Calculate the load supported by each man(3 marks)

SERIES 4

SECTION A (30 MARKS)

Answer ALL questions in this section.

1. Moraa, a Form 10 learner at Watamu Senior School, attends a career guidance session. During the session, Engineer Omondi explains how various physics branches are applied to solve real-life problems in the community.

(a) Define the following term he explained: (1 mark)

Physics

____

____

(b) Match each activity carried out by Engineer Omondi with the correct branch of Physics: (3 marks)

Task Performed	Branch of Physics
(i) Designing a refrigeration system for a dairy plant in Narok	(a) Geometrical Optics
(ii) Determining the force required to pull a tractor stuck in mud	(b) Thermodynamics
(iii) Repairing a lighthouse lens along the Coast	(c) Mechanics

(c) The engineer mentions that a radiologist applies Physics in hospitals.
State the specific branch of Physics involved. (1 mark)

___________________________________.

(d) Explain one connection between Physics and Geography . (1 mark)

____

____

2. Chief Lemayian is vaccinating cattle using a syringe. He observes that when he pulls the plunger, the liquid enters the syringe.

(a) Using the concept of atmospheric pressure, explain why the liquid flows into the syringe. (1 mark)

____

____

(b) The Chief's son drinks fermented milk using a straw.
If pressure at the bottom of the glass is 101,500 Pa and atmospheric pressure is 101,000 Pa, determine the pressure due to the liquid only. (2 marks)

(c) Explain why heavy tractors are fitted with wide tyres instead of narrow ones. (1 mark)

____

____

3. Achieng uses a slingshot made of a rubber band. She notices that excessive stretching causes it to snap permanently.

(a) Name the point beyond which the material does not return to its original shape. (1 mark)

____.

(b) A spring stretches by 2 cm when a force of 5 N is applied.
Calculate the spring constant (k). (2 marks)

(c) Differentiate between: (2 marks)

i. Ductility
ii. Brittleness

Ductility	Brittleness

4. An electric iron uses a bimetallic strip to control temperature in a modern house in Nanyuki Town.

(a) Name the two metals typically used in the strip. (2 marks)

i. ____.
ii. ____.

(b) Explain how bending of the strip prevents overheating. (2 marks)

____

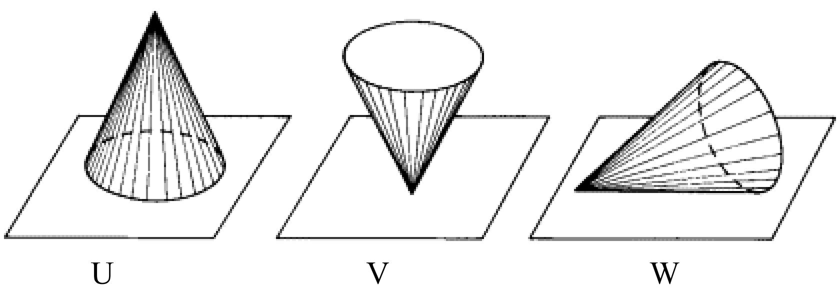
____

(c) Give one reason why overhead power lines are left slack between poles. (1 mark)

____

____

5. Wekesa, a paper artist, is constructing a birthday cap for his children and wants to ensure it is stable.



(a) From the diagram provided, identify the most stable cap and explain why. (2 marks)

____

____

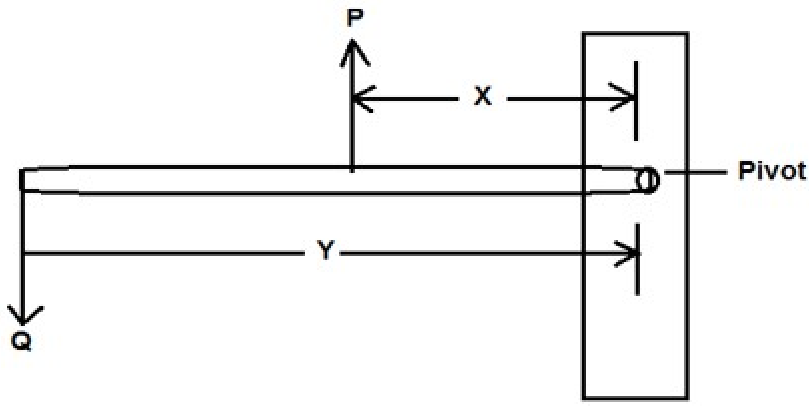
____

(b) Define moment of a force. (2 marks)

____

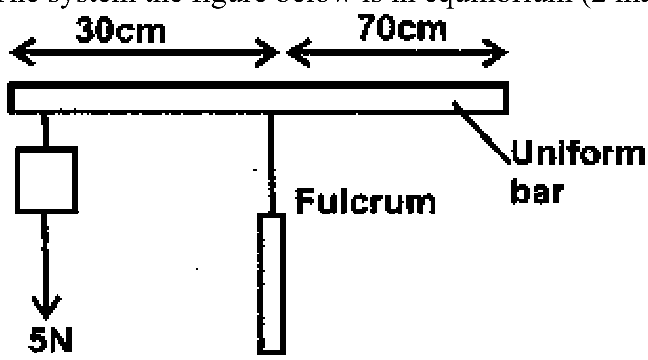
____

(c) The figure below shows a beam of negligible weight balanced by constant forces P and R.



Derive the relationship between X and Y (2mks)

(d) The system the figure below is in equilibrium (2 marks)



Determine the weight of the bar.

(e)

(f) State the two conditions necessary for a body to be in equilibrium: (2 marks)

(i) ____.

(ii) ____.

SECTION B (60 MARKS)

6. Mutua is lifting a crate weighing 800 N to a height of 5 m using a pulley system in 20 seconds. Velocity Ratio = 2

(a) Calculate the work done. (2 marks)

(b) Determine the power produced. (2 marks)

(c) If the applied effort is 500 N, calculate the mechanical advantage. (2 marks)

(d) Find the efficiency of the system. (2 marks)

(e) Explain where energy losses occur in the machine. (2 marks)

(i) ____.
(ii) ____.

7. At a music festival, Otieno plays a guitar while Akinyi plays a flute, producing sound waves.

(a) State the type of wave produced by a vibrating guitar string. (1 mark)

____.

(b) Differentiate between: (2 marks)

i. Reflection of sound
ii. Refraction of sound

(c) Explain resonance and its role in amplifying sound. (2 marks)

(d) Calculate the wavelength of a sound wave with frequency 170 Hz and speed 340 m/s. (2 marks)

8. At Mary Hill Girls Senior School, Grade 10 learners are carrying out a Physics experiment in the laboratory to study wave behavior. Under the guidance of their teacher, they explore interference and diffraction patterns using light and simple apparatus.

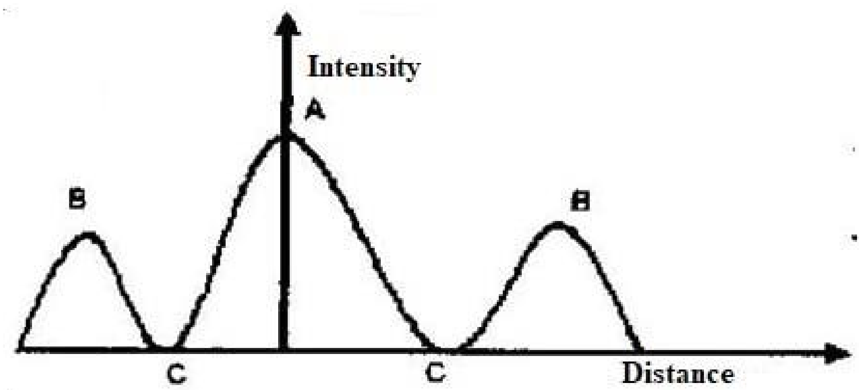
a. Distinguish between stationary and progressive waves. (2 marks)

b.

i. Describe how a young's double slit may be made in a laboratory. (2 marks)

ii. State the condition for a minimum to occur in an interference pattern. (2 marks)

c. The sketch graph below shows the results of an experiment to study diffraction patterns using double silt.



i. Sketch an experimental set up that may be used to obtain such a pattern. (2 marks)

ii. Name an instrument for measuring intensity (1 mark)

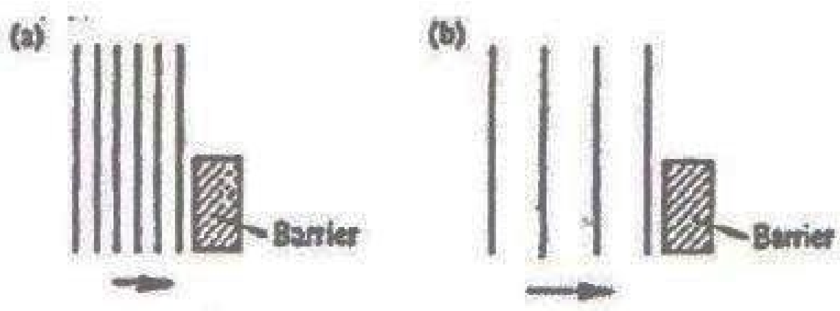
____.

iii. Explain how the peaks labelled A and B and troughs labeled C are formed. (2 marks)

____

____

9. Figures (a) and (b) below show wave fronts incident on barriers blocking part of the path.



On the same figures sketch the wave fronts to show the behaviour of the waves as they pass each barrier and after passing the barrier. (1mk)

10. Scientists use radioactive tracers to study underground water movement.

a. What is meant by the following terms: (2 marks)

i. Radioactive decay

____

____

ii. isotope.

____

____

b. The table shows how the activity (disintegrations per minute) of a sample of carbon-14 varies with time (in years).

Time (grs)	0	2500	5000	7500	10000	12500	17250	20000
Disintegration/min	15	11	8	5	4.0	3.2	1.6	1.2

i. Plot a graph of activity against time (x-axis). (2 marks)

ii. Estimate the half-life of carbon-14 from the graph. (2 marks)

c.

i. Draw a labeled diagram of a Geiger- Muller tube. (2 marks)

ii. Explain how it detects radioactive particles/rays. (1 mark)

d. State one use of radioactivity in each of the following;

i. Medicine(1 mark)

ii. Agriculture. (1 mark)

iii. Radon gas $^{222}{}_{86}Rn$ decays by emission of α particles. Show by use of an equation the transformation of the gas. (2 marks)

iv. Give two uses of cobalt - 60 as a Radioactive source. (2 marks)

11. (a) A radioactive carbon -14 decays to Nitrogen by beta emissions as shown.

$$^{14}_{6}C \rightarrow {}^{x}_{7}N + {}^{0}_{y}\beta$$

Determine the values of x and y in the equation. (2 marks)

b. State two safety precautions when disposing radioactive waste. (2 marks)

12. At Precious Blood Riruta Senior School, Grade 10 learners are conducting a Physics practical to study heat and changes of state in the laboratory.

a. Define specific latent heat of fusion (1mk)

b. You are provided with the following apparatus:

A filter funnel, a thermometer, a stop watch, ice at 0°C, an immersion heater rated P watts, a beaker, a stand, boss and clamp and a weighing machine. Describe an experiment to determine the specific latent heat of fusion of ice. Clearly state the measurements to be made. (3mks)

c. 200 g of ice at 0°C is added to 400 g water in a well lagged calorimeter of mass 40 g.The initial temperature of the water was 40°C. If the final temperature of the mixture is X°C,

(*Specific latent of fusion of ice L = 3.36 x 105 Jkg^{-1}, specific heat capacity of water, c = 4200 $Jkg^{-1}K^{-1}$, specific heat capacity of copper = 400 $Jkg^{-1}K^{-1}$*)

i. Derive an expression for the amount of heat gained by ice to melt it and raise its temperature to X°C (2mks)

ii. Derive an expression for the amount of heat lost by the calorimeter and its content when their temperature falls to X°C. (2mks)

iii. Determine the value of X. (2mks)

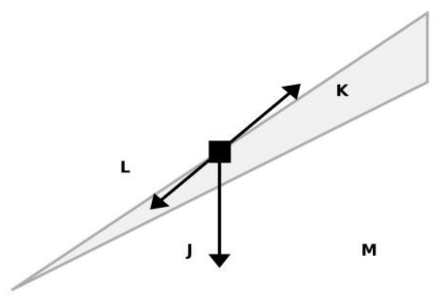
SERIES 5

1. A car accelerates uniformly from rest to a velocity of 25 m/s in 8 seconds. Calculate:

   (a) The acceleration of the car

   (b) The distance covered during this acceleration (4 marks)

2. Study the diagram below showing forces acting on an object on an inclined plane:



   (a) Identify the forces labeled J, K, and L

   (b) What is represented by angle M?

   (c) If the mass of the object is 5 kg and the angle is 30°, calculate the component of weight parallel to the plane.

(6 marks)

3. A 2 kg object falls freely from a height of 10 meters.

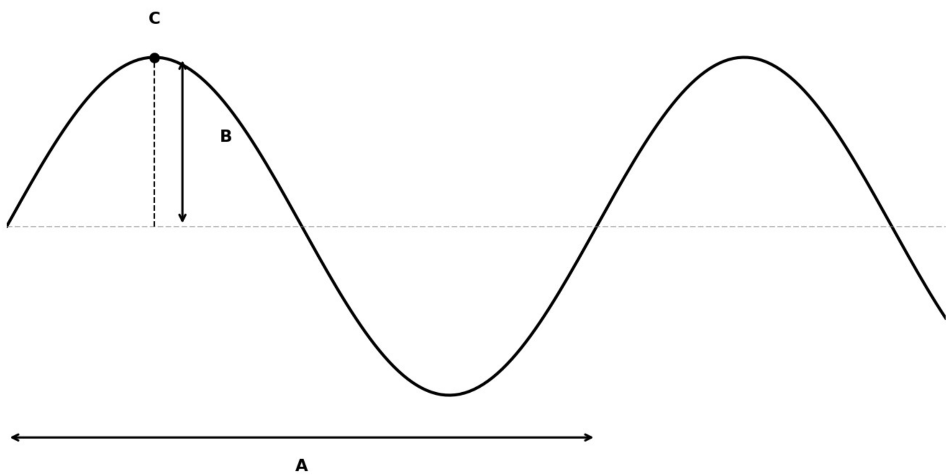
(a) Calculate its potential energy at the initial height.

(b) Calculate its kinetic energy just before hitting the ground.

(c) What principle does this demonstrate?(5 marks)

4. Distinguish between elastic and inelastic collisions(3 marks)

5. The diagram below shows a transverse wave:

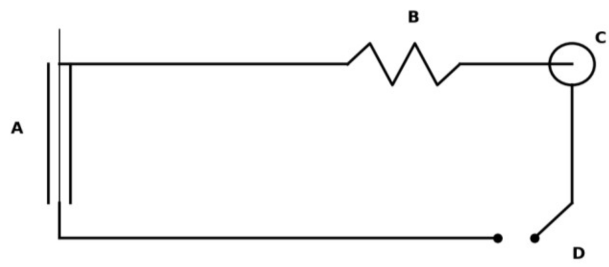


(a) Label the parts identified as A, B, and C on the diagram.

(b) If the frequency of the wave is 2 Hz and wavelength is 6.28 m, calculate the wave speed.

(c) State the difference between a longitudinal and transverse wave. (6 marks)

6. Study the circuit diagram below:



(a) Name the components labeled A, B, C, and D in the circuit.

(b) State the function of component D.

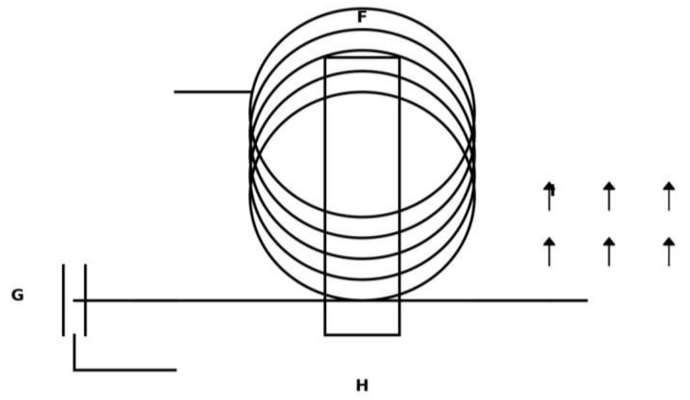
(c) If the EMF of the battery is 12 V and the total resistance is 4 Ω, calculate the current in the circuit using Ohm's law(7 marks)

7. (a) State Kirchhoff's current law.

(b) Two resistors of 3 Ω and 6 Ω are connected in parallel across a 12 V supply. Calculate the total resistance.

(c) Calculate the total current supplied by the battery. (7 marks)

8. The diagram below shows an electromagnet setup:



(a) Identify the components labeled F, G, H, and I.

(b) What is the function of component H in the electromagnet?

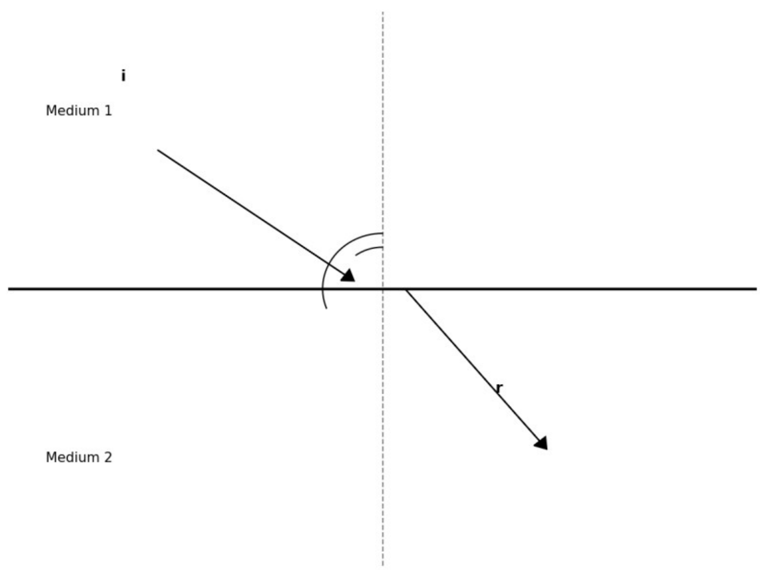
(c) How would increasing the number of turns in the coil affect the strength of the electromagnet

(6 marks)

9. (a) State Fleming's left-hand rule.

(b) A current-carrying conductor of length 0.5 m is placed in a magnetic field of 0.8 T. If the current is 2 A, calculate the force on the conductor. (5 marks)

10. The diagram below shows light refraction at an interface between two media:

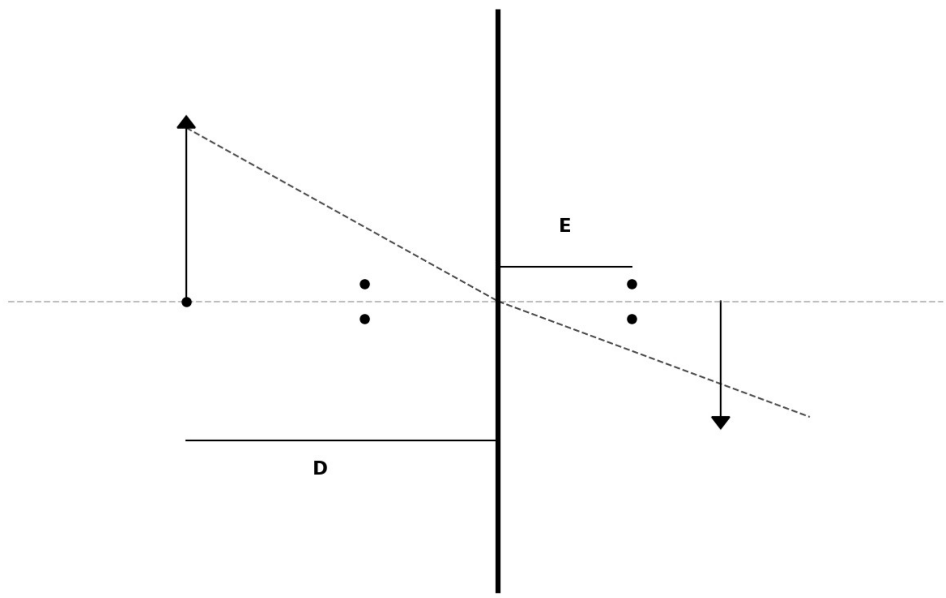


(a) Identify the angles labeled i and r on the diagram.

(b) State Snell's law of refraction.

(c) If the angle of incidence is 30° and the refractive index of medium 2 is 1.5, and medium 1 is air (n=1), calculate the angle of refraction.(7 marks)

11. The diagram below shows a convex lens with an object placed in front of it:



(a) Identify the points and distances labeled D and E on the diagram.

(b) Using the lens formula 1/f = 1/u + 1/v, where u = 30 cm and f = 10 cm, calculate the image distance (v).

(c) Calculate the magnification of the image. (8marks)

SERIES 6

INSTRUCTIONS
ANSWER ALL QUESTIONS IN THE SPACES PROVIDED AFTER EACH QUESTION.

1. State two branches of physics. (2marks)

2. State any three subjects that relate to physics and how they do relate? (3 marks)

3. State three careers that are physics dependent. (3 marks)

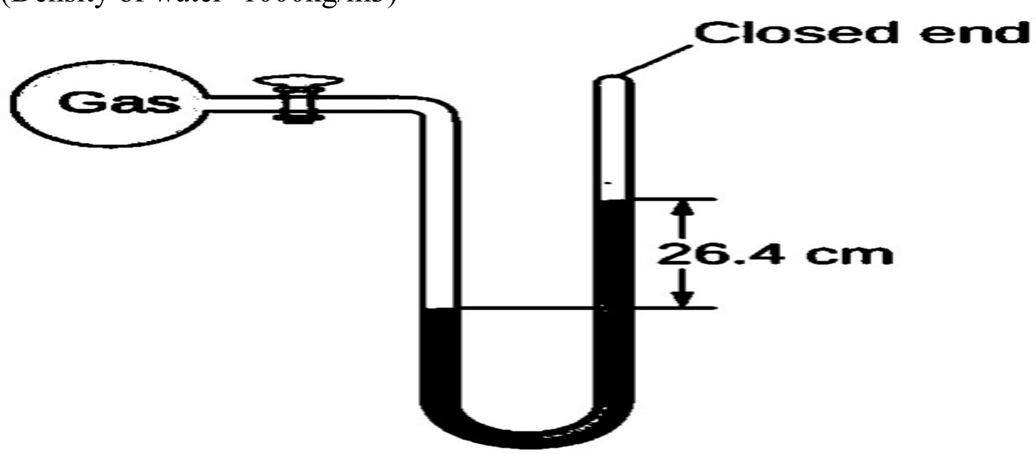
4. A balloon rising in the atmosphere may burst as it rises. Explain this phenomena. (2 marks)

5. State two factors that affect pressure in fluids. (2 marks)

6. Atmospheric pressure is crucial in daily life. Explain an experiment you would carry out to demonstrate its existence. (5 marks)

7. When measuring pressure SI units are used state its SI units. (1 mark)

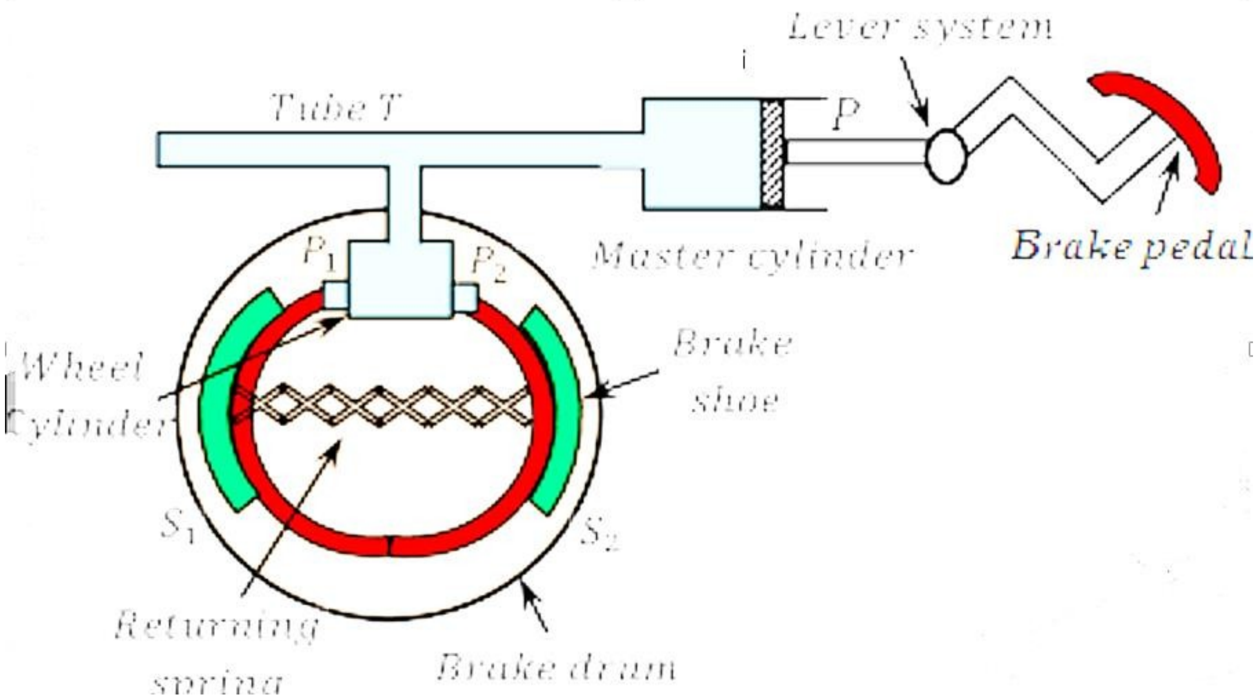
8. Determine the pressure of the gas below that raises the level of water as shown below.(3 marks)
(Density of water=1000kg/m3)



9. A fisherman dives 25m below the sea water. Given that the sea water is 1.03g/cm3, gravitational force (g) is 10N/Kg and atmospheric pressure is 103, 000n/m2. Determine the following:
   i) The pressure exerted by the sea water on the man. (3 marks)

   ii) Total pressure on him. (2 marks)

10. Below is a diagram of a hydraulic break system explain how the system works (4 marks)



11. A student was given a pipe to irrigate a land from a tank full of water with no tap and opened on the upper part raised 10m high from the ground. Explain as a physics student how you would irrigate the land. (3 marks)

12. Define the following term relating to material properties;
    a. Elastic deformation. (2 marks)

b. Plastic deformation. (2 marks)

c. Ductility (2 marks)
d. Malleability (2 marks)

e. Brittleness (2 marks)

13. State Hooke's law (1 mark)

14. A spring produces an extension of 12mm when a force of 0.6N is applied to it. Calculate the spring constant for the system that has two such springs arranged in:
    a) Series (3 marks)

    b) Parallel (3 marks)

15. A wire of length 2 m and cross sectional area of 0.0001m ^{2} is stretched by a load of 1020 N. The wire is stretched by 0.1 cm. Calculate the following:
    a. Longitudinal or tensile stress (3 marks)

    b. Longitudinal or tensile strain (3 marks)

    c. Young's modulus of elasticity (3 marks)

16. Explain how a force pump works (5 marks)

17. What are the advantages of a force pump over a lift pump (4 marks)

18. As a physics student how can you use your knowledge to better your community (6 marks)

SERIES 7

Answer all questions

1. A student is exploring career opportunities in science and wants to understand how Physics relates to other fields.

a) i) Define Physics as a branch of science. (1 mark)

____

____

ii) List **two branches of Physics**. (1 mark)

_________________________.

_________________________.

iii) Give **two examples of how Physics is applied in day-to-day life**. (2 marks)

____

____

iv) Mention **two careers related to Physics**. (2 marks)

____________________________.

____________________________.

2. **a)** i) Define **atmospheric pressure**. (2 marks)

____

____

ii) State **two factors affecting pressure in fluids**. (2 marks)

____

____

iii) State **one application of fluid pressure in daily life**. (1 mark)

____

____

b) i) calculate the pressure at the bottom of a tank containing water 1.5 m deep. (Take $\rho = 1000\ kg/m^3$ $g = 10\ N/kg$). (5 marks)

ii) Explain **the principle of transmission of pressure in fluids** and give one example of its application. (5 marks)

____

vi) Using a well labelled diagram Describe an experiment to **demonstrate atmospheric pressure using a glass and water**. (5 marks)

3. A student wants to design a spring system for a small machine.

a) i) Define **elasticity** and **ductility**. (2 marks)

Elasticity:

Ductility:

ii) State **two other mechanical properties of materials**. (2 marks)

b) A wire of original length L_0=2.0 m stretches by ΔL=4 mm under a force of F=200 N and cross-sectional area A=2×10−6 m^2 Calculate:

i. Tensile stress (σ=F/A) (2 marks)

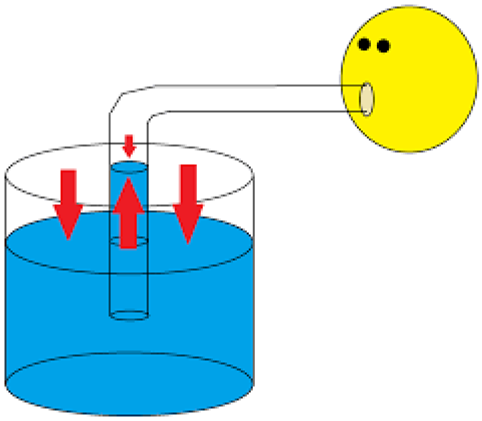
iv) State two **applications of elasticity of materials in industry and daily life**. (4 marks)

4. a) i) Give **two examples of how Physics is related to other sciences**. (2 marks)
physics and chemistry:

Physics and biology:

ii) Explain briefly **the importance of Physics in technological innovations.** (2 marks)

5. During a science demonstration, a student observes a drinking straw in a glass of water.



a) i) Explain why water rises in the straw when pressure is applied. (2 marks)

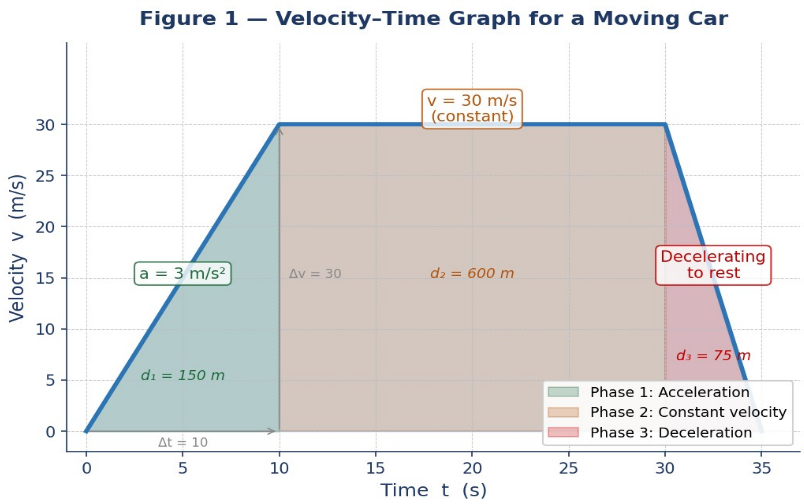
ii) Name **two other applications of atmospheric pressure**. (2 marks)

b) i) using a well labelled diagram show an experiment to show **air pressure can support a card on a glass of water when inverted**. (3 marks)

ii) Explain **the effect of atmospheric pressure on human body in high-altitude conditions**. (3 marks)

SERIES 8

1.A car starts from rest and accelerates uniformly to a velocity of 30 m/s in 10 seconds. It then travels at this constant velocity for 20 seconds before decelerating uniformly to rest in 5 seconds.



(a) Define the term acceleration and state its SI unit. (2 marks)

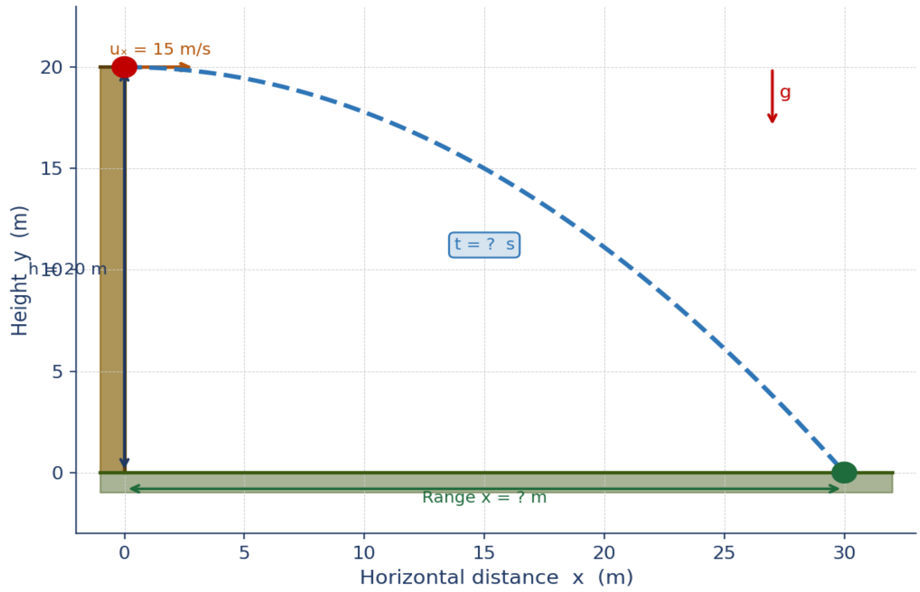
(b) Calculate the acceleration of the car during the first phase of motion. (3 marks)

(c) Using Figure 1, calculate the total distance travelled by the car. (4 marks)

(d) State one real-life application of uniform deceleration. (1 mark)

2.A ball is kicked horizontally from the top of a cliff 20 m high with an initial horizontal velocity of 15 m/s. ($g = 10\ m/s^2$)

Figure 2 — Projectile Motion from a Cliff



(a) Distinguish between a scalar quantity and a vector quantity, giving one example of each. (2 marks)

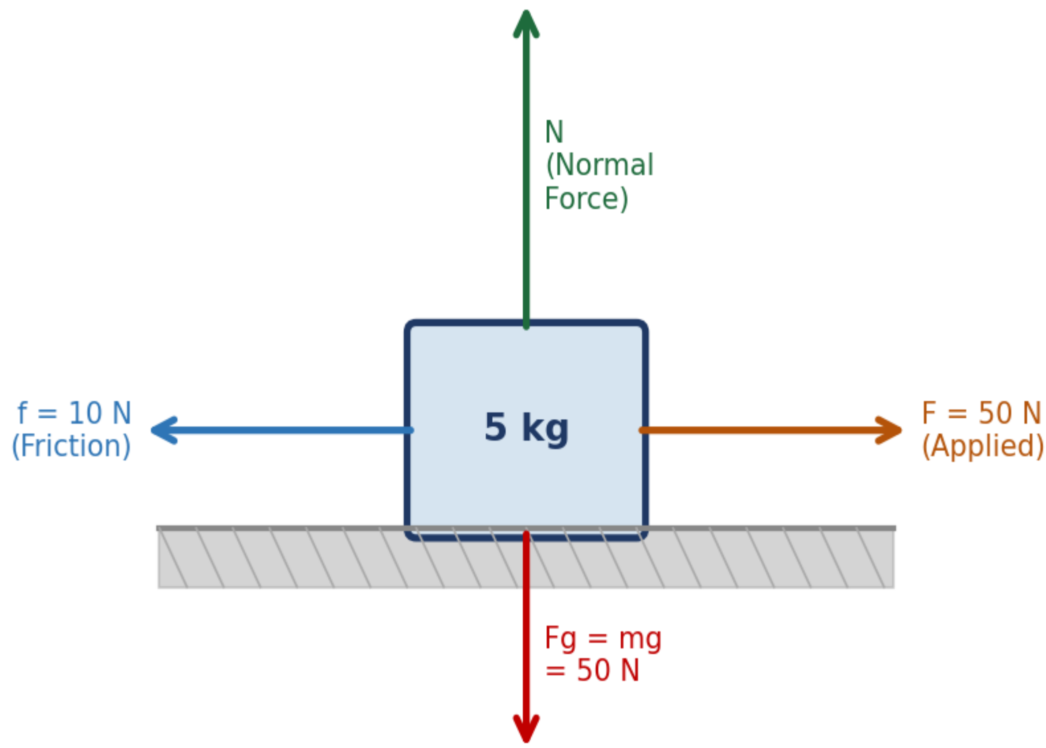
(b) Calculate the time taken for the ball to reach the ground. (3 marks)

(c) Determine the horizontal range covered before the ball hits the ground. (3 marks)

(d) State the principle that explains why the horizontal and vertical motions are independent. (2 marks)

3.Study the free body diagram below carefully and answer the questions that follow.

Figure 3 — Free Body Diagram of 5 kg Block



(a) State Newton's First Law of Motion. (2 marks)

(b) Calculate the net force acting on the block. (2 marks)

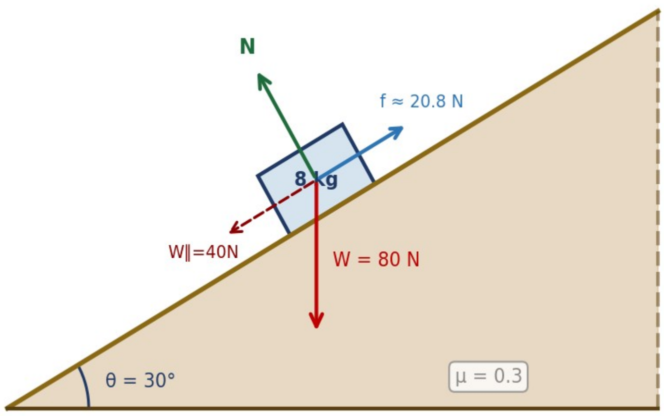
(c) Using Newton's Second Law, calculate the acceleration of the block. (3 marks)

(d) State one practical application of Newton's Third Law of Motion. (1 mark)

(e) Define the term 'inertia' and state what it depends on. (2 marks)

4.A wooden block of mass 8 kg is placed on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction μ = 0.3.

Figure 4 — Block on Inclined Plane (θ = 30°)



(a) Resolve the weight of the block into components parallel and perpendicular to the incline. (3 marks)

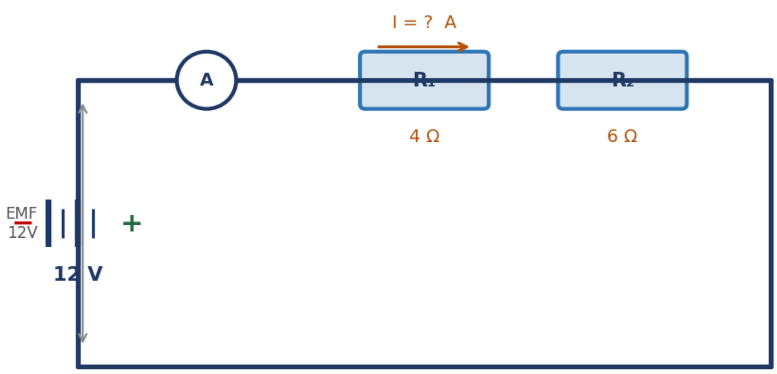
(b) Calculate the frictional force acting on the block. (2 marks)

(c) Determine whether the block will slide down the plane. Explain your answer. (3 marks)

(d) State two factors that affect the magnitude of the frictional force between two surfaces. (2 marks)

5.Study the series circuit diagram below and answer the questions that follow.

Figure 5 — Series Circuit: R_1 = 4 Ω and R_2 = 6 Ω with 12 V Battery



(a) State Ohm's Law. (2 marks)

(b) Calculate the total resistance in the series circuit. (2 marks)

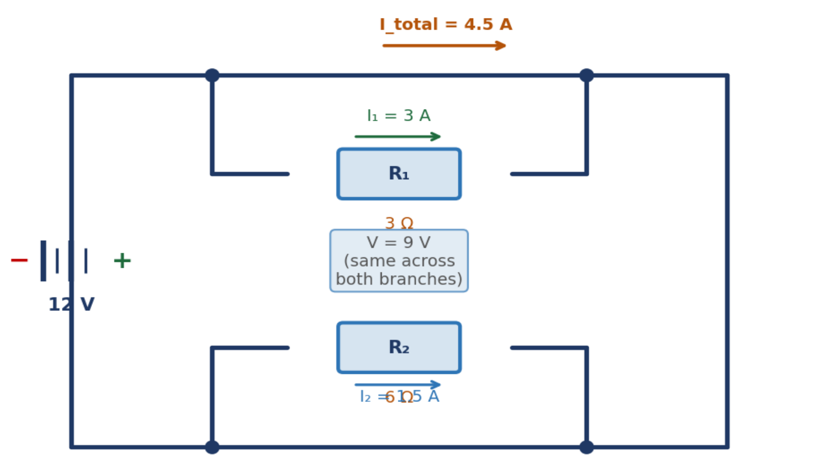
(c) Calculate the current flowing through the circuit. (2 marks)

(d) Calculate the potential difference (voltage drop) across each resistor. (2 marks)

(e) State one difference between series and parallel circuits in terms of current flow. (2 marks)

6.The circuit below shows two resistors connected in parallel to a 9 V battery.

Figure 6 — Parallel Circuit: R_1 = 3 Ω and R_2 = 6 Ω with 9 V Battery



(a) Calculate the equivalent resistance of the parallel combination. (3 marks)

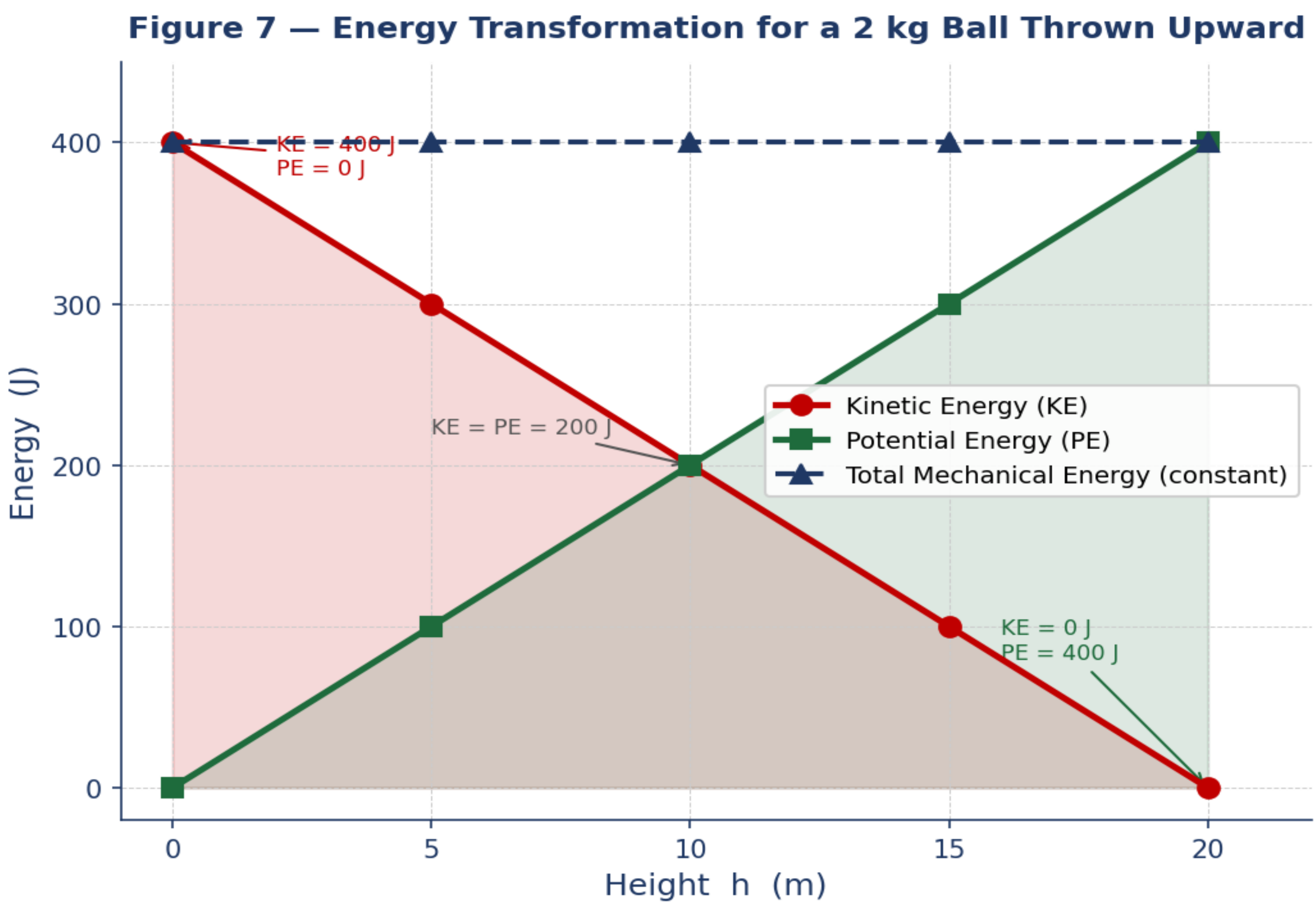
(b) Calculate the total current drawn from the battery. (2 marks)

(c) Calculate the electrical power dissipated in resistor R_1. (2 marks)

(d) State two advantages of connecting household appliances in parallel. (2 marks)

(e) Define electric current and state its SI unit. (1 mark)

7.A 2 kg ball is thrown vertically upward from the ground with an initial speed of 20 m/s.



(a) Define work done and state its SI unit. (2 marks)

(b) Calculate the initial kinetic energy of the ball. (2 marks)

(c) Using conservation of energy, calculate the maximum height reached by the ball. (3 marks)

(d) Complete the table below using Figure 7 as a guide: (2 marks)

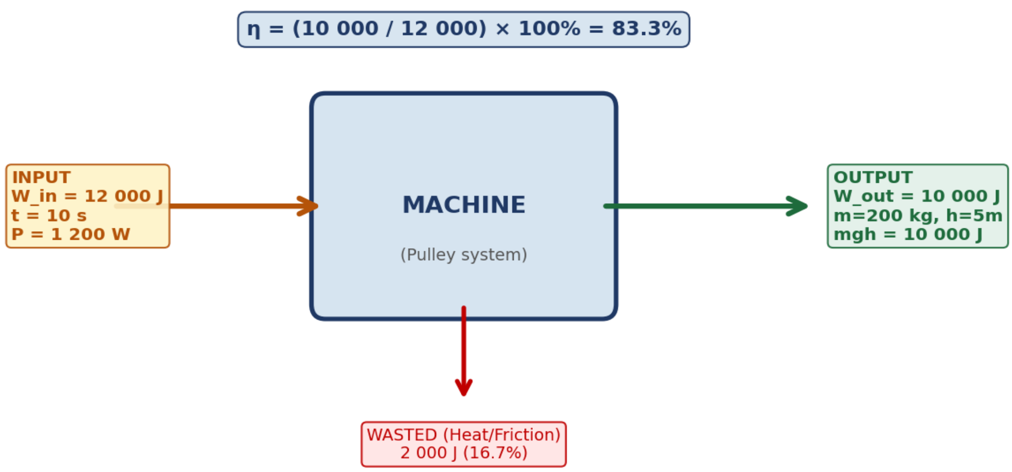
Position	Kinetic Energy (J)	Potential Energy (J)	Total Mechanical Energy (J)

Ground level	400 J	0 J	
Half-way up			400 J
Max. height	0 J		400 J

(e) State the Law of Conservation of Energy. (1 mark)

8.A machine is used to lift a 200 kg load through a height of 5 m in 10 seconds. The machine requires an input energy of 12 000 J.

Figure 8 — Energy Flow Diagram for a Machine



(a) Define power and state its SI unit. (2 marks)

(b) Calculate the useful work output done by the machine. (2 marks)

(c) Calculate the efficiency of the machine. (2 marks)

(d) Calculate the power input of the machine. (2 marks)

(e) State two ways in which the efficiency of a machine can be improved. (2 marks)

SERIES 9

SECTION A: (40 Marks)

Answer all questions in this section.

1. A student is exploring career opportunities in science and wants to understand how Physics relates to other fields.

a)
i) Define Physics as a branch of science. (1 mark)

____

____

ii) List **two branches of Physics**. (1 mark)
____.

____.

iii) Give **two examples of how Physics is applied in day-to-day life**. (2 marks)
____

____

iv) Mention **two careers related to Physics**. (2 marks)

____.

____.

2. A syringe is used to demonstrate how pressure is transmitted in liquids.

a)
i) Define **atmospheric pressure**. (2 marks)
____

____

ii) State **two factors affecting pressure in fluids**. (2 marks)
____

____

iii) State **one application of fluid pressure in daily life**. (1 mark)

____

____

b)
i) calculate the pressure at the bottom of a tank containing water 1.5 m deep. (Take ρ=1000 kg/m^3 g=10 N/kg). (5 marks)

ii) Explain **the principle of transmission of pressure in fluids** and give one example of its application. (5 marks)

vi) Using a well labelled diagram Describe an experiment to **demonstrate atmospheric pressure using a glass and water**. (5 marks)

3. A student wants to design a spring system for a small machine.

a)

i) Define **elasticity** and **ductility**. (2 marks)

elasticity:

Ductility:

ii) State **two other mechanical properties of materials**. (2 marks)

b) A wire of original length L_0=2.0 m stretches by ΔL=4 mm under a force of F=200 N and cross-sectional area A=2×10−6 m² Calculate:

i. Tensile stress (σ=F/A) (2 marks)

ii. Tensile strain (ϵ=ΔL/L0) (2 marks)

iii. Young's modulus (Y=σ/ϵY) (2 marks)

iv) State two **applications of elasticity of materials in industry and daily life**. (4 marks)

v) Using a well labelled diagram, describe **a simple experiment to demonstrate Hooke's Law using a spring and weights**. (4 marks)

4. a)

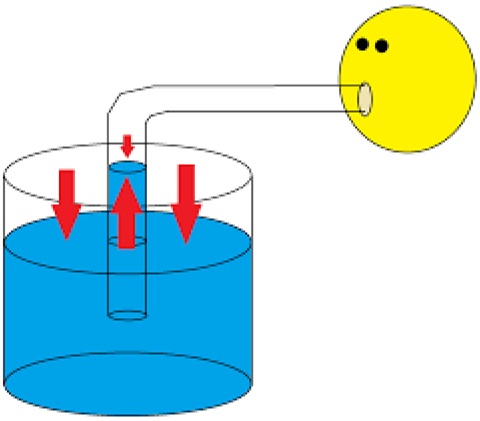
i) Give **two examples of how Physics is related to other sciences**. (2 marks)

physics and chemistry:

Physics and biology:

ii) Explain briefly **the importance of Physics in technological innovations**. (2 marks)

5. During a science demonstration, a student observes a drinking straw in a glass of water.



a)

i) Explain why water rises in the straw when pressure is applied. (2 marks)

ii) Name **two other applications of atmospheric pressure**. (2 marks)

b)

i)using a well labelled diagram show an experiment to show **air pressure can support a card on a glass of water when inverted**. (3 marks)

ii) Explain **the effect of atmospheric pressure on human body in high-altitude conditions**. (3 marks)

SERIES 10

SECTION A: (40 Marks)

Answer all questions in this section.

1. While demonstrating a science experiment during a school open day, a visitor asks *"What exactly is Physics and why do you study it?"*
a) Define Physics. (1 mark)

____

____

b) State two reasons why Physics is important in society. (2 marks)

i. ____
ii. ____

2. A student found the following table in an old physics textbook but the branches were mixed up. Match the branches in List A with the correct description in List B. Write the correct letter in the space provided. (4 marks)

List A: Branch of Physics	List B: Description
i. Mechanics	A. Study of heat and temperature
ii. Thermodynamics	B. Study of motion of bodies
iii. Optics	C. Study of sound and its properties
iv. Acoustics	D. Study of light

3. A team of experts is designing a new hospital theatre.
Explain how Physics relates to each of the following fields:
a) Biology (1 mark)

____

____

b) Medicine (1 mark)

____

____

c) Engineering (1 mark)

____

____

d) Agriculture (1 mark)

____

____

4. A Career Day is organised in your school. Mention three career opportunities directly related to Physics that a visiting speaker may talk about. (3 marks)

i. ______________________________

ii. ______________________________

iii. ______________________________

5. A cook presses a sharp knife and a blunt knife on a tomato. The sharp one cuts easily while the blunt one does not.
a) Define pressure (1 mark)

____

____

b) Explain why the sharp knife cuts more easily. (2 marks)

____

____

c) State the SI unit of pressure. (1 mark)

____

____

6. A diver descends deeper into the ocean and notices his ears feel more pressure.
a) State two factors that affect pressure in liquids. (2 marks)

i. ______________________________

ii. ______________________________

b) Explain why the diver feels more pressure as he goes deeper. (2 marks)

____

____

7. A soft plastic bottle is sealed tightly at the top of Mount Kenya and later opened at sea level. The bottle collapses inward immediately.
a) Define atmospheric pressure. (1 mark)

____

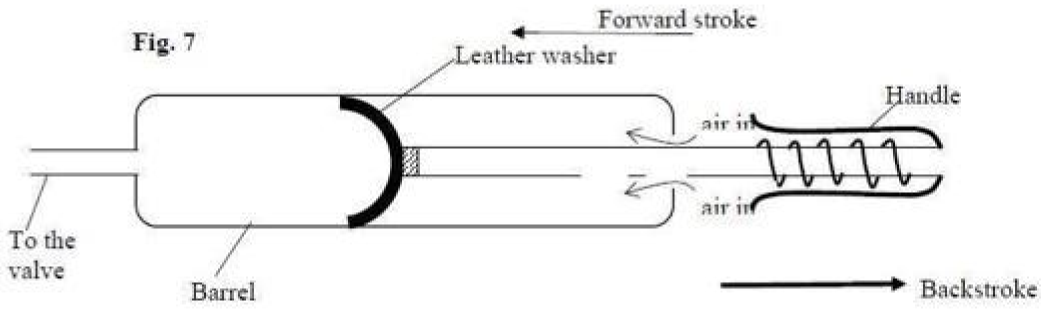
____

b) Explain why the bottle collapses at sea level. (2 marks)

____

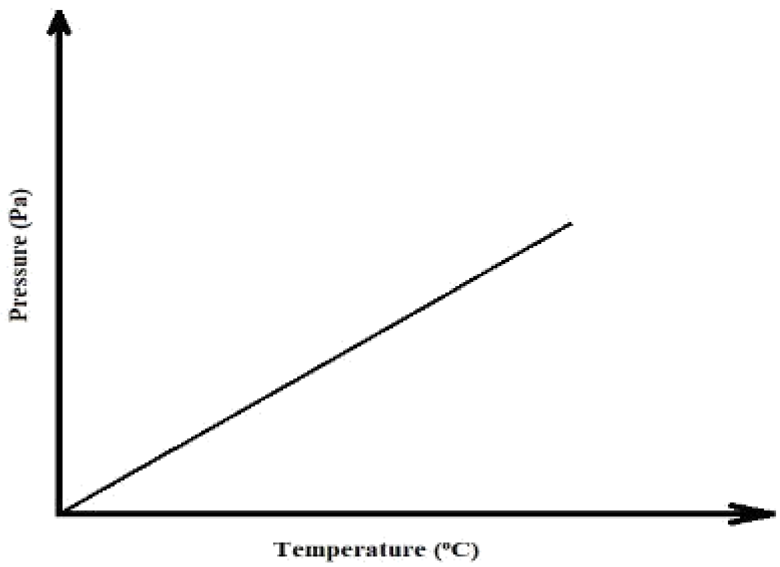
____

c) The bicycle pump shown in Figure 7 below is one of the applications of pressure in gases.



Explain how the pump operates during backstroke. (2mks)

d) The sketch graph below shows the relationship between pressure and temperature of a gas in a fixed volume container.



State the relationship between pressure and temperature that can be deduced from the graph (1 Mark)

8. a) A rectangular block of mass 8 kg rests on a table. Its base area is 0.02 m^2.
Calculate the pressure it exerts on the table. (3 marks)

b) A liquid of density 1000 kg/m^3 is filled in a tank to a depth of 5 m.
Calculate the pressure at the bottom of the tank using $P = \rho gh$.
(Take $g = 9.8\ m/s^2$). (3 marks)

c) Given that density of water = 1000Kgm^{-3}, density of kerosene = 800Kgm-3 and atmospheric pressure = 1.03×10^5 pascals, determine the pressure acting on the base of the container

SECTION B (60 MARKS)

Answer all questions from this section.

9. A new borehole water supply system is being installed in a village dispensary. The technician uses devices that rely on transmission of pressure in fluids.

a) State Pascal's Principle. (2 marks)

b) Using a labelled diagram, explain how a hydraulic lift works in raising a car in a garage. (6 marks)

c) A hydraulic lift has a small piston of area 0.01 m^2 and a large piston of area 0.5 m^2.

If a force of 40 N is applied on the small piston:

i) Calculate the pressure transmitted through the fluid. (3 marks)

ii) Calculate the force exerted on the large piston. (3 marks)

d) Explain how each of the following devices uses pressure differences in fluids:

i) A drinking straw (2 marks)

___

___

ii) A bicycle pump (2 marks)

___

___

iii) A syringe (2 marks)

___

___

10. A science club is testing the use of a syphon to transfer water from an upper tank to a lower tank without using electricity.

a) A syphon tube is used to drain water from a dam. If the height difference between the dam water level and the outlet is 2.5 m, calculate the pressure difference using $P = \rho gh$.

(Take $\rho = 1000\ kg/m^3$, $g = 9.8\ m/s^2$). (2 marks)

b) A tractor tyre has an internal surface area of $0.8\ m^2$. The pressure inside is 2.5×10^5 Pa.

Calculate the force the air exerts on the tyre walls. (2 marks)

c) Explain the working principle of each:

i) Hydraulic brake system in a car (2 marks)

____

____

____

____

c) A fish tank at a pet shop is 0.6 m deep. The shop owner places sensors to detect pressure at different levels.

Explain why pressure in liquids increases with depth. (2 marks)

____

____

____

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d) Water of density 1000 kg/m³ fills the tank.

i) Calculate the pressure at a depth of 0.2 m. (2 marks)

ii) Calculate the pressure at the bottom of the tank. (2 marks)

e) Explain the effect of increasing each of the following on the pressure at the bottom of the tank:

i) Density of the liquid (2 marks)

ii) Depth of the liquid (2 marks)

f) The tank has a base area of 0.3 m^2.

Calculate the total force exerted on the base by the water at full capacity. (2 marks)

g) Explain why deep-sea animals require stronger body structures compared to surface animals. (2 marks)